NAME-UTSAV MISHRA

REG NO-24BAI10564

Slot: B22
Submission Date: 30/03/26

OPEN SOURCE AUDIT PROJECT – GIT

1. Introduction

Open Source Software (OSS) plays a crucial role in modern computing by allowing users to freely access, modify, and distribute software. It promotes transparency, collaboration, and innovation. This project focuses on performing an audit of an opensource software, analyzing its features, license, philosophy, and real-world impact.

2. About the Selected Software

The selected open-source software for this project is Git. Git is a distributed version control system developed by Linus Torvalds in 2005. It is widely used by developers to track changes in source code and manage collaborative software development projects.

Git allows multiple developers to work on a project simultaneously without conflicts, making it an essential tool in modern software engineering.

3. Features of Git

Git provides several powerful features:

Distributed Version Control: Every user has a complete copy of the repository
Branching and Merging: Enables parallel development
Fast Performance: Efficient handling of large projects
Data Integrity: Uses hashing to ensure data consistency
Collaboration: Supports teamwork through platforms like GitHub

4. License Analysis

Git is licensed under the GNU General Public License (GPL).

This license provides the following freedoms:

Freedom to use the software for any purpose
Freedom to study and modify the source code
Freedom to distribute copies
Freedom to distribute modified versions

The GPL ensures that the software remains open and accessible to everyone.

5. Open Source Philosophy

Git reflects the core principles of open-source philosophy:

Transparency: Source code is openly available
Collaboration: Developers worldwide contribute
Freedom: Users can modify and share software
Community-driven development: Improvements come from contributors 6. Impact on the Real World

Git has revolutionized software development practices. It is used by millions of developers and organizations worldwide. Platforms like GitHub rely on Git for version control and collaboration.

It has enabled:

Efficient team collaboration
Faster development cycles
Better project management
7. Practical Implementation

As part of this project, Git was used to perform basic operations.

Commands Used:
git init git
add .
git commit -m "Initial commit"

These commands initialize a repository, track files, and save changes.

8. Shell Scripts Implementation

Five shell scripts were created and executed in a Linux-like environment (Git Bash):

File Counter Script
Counts the number of files in a directory Backup Script
Creates a backup of files with timestamp
System Information Script
Displays user, system, and directory details
File Check Script
Verifies whether a file exists
Calculator Script
Performs basic arithmetic operations

These scripts demonstrate basic Linux command-line and automation skills.

9. Testing and Results

All scripts were successfully executed using Git Bash. The outputs were verified and found correct. Screenshots of execution were taken as proof of implementation.

10. Conclusion

This project provided a comprehensive understanding of open-source software and its importance. Git was studied in detail, including its features, license, and impact. Additionally, practical skills in shell scripting and Linux commands were developed.

The project highlights how open-source tools like Git promote collaboration, transparency, and innovation in the software industry.

Here are some screenshots of the terminal-

```
Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit (master)
$ cd scripts

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ chmod +x *.sh

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./file_counter.sh

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./backup.sh
📦 Creating backup...
✅ Backup created in backup_20260329_214945
```

```
Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./system_info.sh
📊 System Information
---------------------
User:
Hostname: DESKTOP-ONE8P1T
Date: Sun Mar 29 21:50:02 IST 2026
Current Directory: /c/Users/Lenovo/Desktop/open-source-audit/scripts
OS: Msys

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./check_file.sh
Enter filename: README.md
❌ File does not exist

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./calculator.sh
🧮 Simple Calculator
Enter first number: 12
Enter second number: 12
Choose operation (+ - * /):
/
Result: 1
```

```
Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./file_counter.sh
🗁 Counting files in current directory...
Total files: 6

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ ./check_file.sh
Enter filename: ../README.md
☑ File exists

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ git init
Initialized empty Git repository in C:/Users/Lenovo/Desktop/open-source-audit/scripts/.git/

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ git add .
warning: in the working copy of 'backup.sh', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'backup_20260329_214945/backup.sh', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'backup_20260329_214945/calculator.sh', LF will be replaced by CRLF the
 next time Git touches it
warning: in the working copy of 'backup_20260329_214945/check_file.sh', LF will be replaced by CRLF the
 next time Git touches it
warning: in the working copy of 'backup_20260329_214945/system_info.sh', LF will be replaced by CRLF th
e next time Git touches it
warning: in the working copy of 'calculator.sh', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'check_file.sh', LF will be replaced by CRLF the next time Git touches
it
warning: in the working copy of 'file_counter.sh', LF will be replaced by CRLF the next time Git touche
s it
warning: in the working copy of 'system_info.sh', LF will be replaced by CRLF the next time Git touches
 it

Lenovo@DESKTOP-ONE8P1T MINGW64 ~/Desktop/open-source-audit/scripts (master)
$ git commit -m "Intial commit"
[master (root-commit) 6553d8c] Intial commit
 10 files changed, 94 insertions(+)
 create mode 100644 backup.sh
 create mode 100644 backup_20260329_214945/backup.sh
 create mode 100644 backup_20260329_214945/calculator.sh
 create mode 100644 backup_20260329_214945/check_file.sh
 create mode 100644 backup_20260329_214945/file_counter.sh
 create mode 100644 backup_20260329_214945/system_info.sh
 create mode 100644 calculator.sh
 create mode 100644 check_file.sh
 create mode 100644 file_counter.sh
 create mode 100644 system_info.sh
```